SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I SOWNDARSHINI M (RE5Ow6) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sowndarshini M

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	2	1	1	1
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

26/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Answer

Introduction

Subnetting is a networking technique used to divide a large IP network into smaller logical networks called subnets. It helps network administrators utilize IP addresses efficiently, reduce unnecessary broadcast traffic, improve network performance, and enhance network management and security.

In a Smart Campus, different academic departments may require separate networks to connect computers, printers, servers, and other devices. By assigning a separate subnet to each department, the administrator can manage network traffic efficiently.

The given network address block is 192.168.100.0/24. The following subnet masks have been allocated to three departments:

- School of Computing – /25
- School of Electronics – /26
- School of Mechanical Engineering – /27

The objective is to determine the network address, broadcast address, valid host range, number of usable host addresses, and remaining unused IP address range for each subnet.

1. Understanding the Given IP Address Block

The given network is:

192.168.100.0/24

The /24 prefix indicates that the first 24 bits represent the network portion of the IP address, while the remaining 8 bits represent the host portion.

The default subnet mask for /24 is:

255.255.255.0

The total number of IP addresses available in this network block is:

$$2^{32-24} = 2^8 = 256$$

Therefore, the complete IP address range is:

192.168.100.0 – 192.168.100.255

These 256 addresses must be divided among the three departments according to their assigned subnet masks.

2. Important Subnetting Formulas

The following formulas are used to calculate the subnet details.

1. Number of total IP addresses

$$\text{Total IPs} = 2^{32-\text{prefix length}}$$

2. Number of usable host addresses

$$\text{Usable hosts} = \text{Total IPs} - 2$$

Two addresses are normally reserved: the network address and the broadcast address.

3. Network address

The network address identifies the subnet. It is the first address in the subnet and cannot normally be assigned to an individual host.

4. Broadcast address

The broadcast address is the last address in the subnet. It is used to send packets to all hosts within that subnet.

5. Valid host range

The valid host range consists of all IP addresses between the network address and the broadcast address.

For this problem, the subnets are assigned in descending order of size, beginning with the lowest available address in the given network block.

3. Subnet Allocation for the School of Computing (/25)

The School of Computing requires a subnet with the prefix length /25.

Step 1: Determine the subnet mask

The subnet mask for /25 is:

255.255.255.128

The number of host bits is:

$$32 - 25 = 7$$

Step 2: Calculate the total number of IP addresses

$$2^7 = 128$$

Therefore, the subnet contains 128 total IP addresses.

Step 3: Determine the network address

The first available address in the given network block is:

Network address: 192.168.100.0

The block size is 128, so the first subnet covers addresses from .0 through .127.

Step 4: Determine the broadcast address

The last address in this subnet is the broadcast address.

Broadcast address: 192.168.100.127

Step 5: Determine the valid host IP address range

The network address and broadcast address cannot be assigned to regular hosts.

Therefore, the valid host range is:

192.168.100.1 – 192.168.100.126

Step 6: Calculate usable host addresses

$$\begin{aligned}\text{Usable hosts} &= 2^7 - 2 \\ &= 128 - 2 \\ &= \boxed{126}\end{aligned}$$

Thus, the School of Computing can connect up to 126 devices using this subnet.

School of Computing – Subnet Details

Network address	192.168.100.0
Subnet mask	255.255.255.128
CIDR prefix	/25
Broadcast address	192.168.100.127
Valid host range	192.168.100.1 – 192.168.100.126
Total IP addresses	128
Usable host addresses	126

4. Subnet Allocation for the School of Electronics (/26)

The School of Electronics requires a subnet with the prefix length /26.

Since the School of Computing uses addresses from 192.168.100.0 to 192.168.100.127, the next available address is 192.168.100.128.

Step 1: Determine the subnet mask

The subnet mask for /26 is:

255.255.255.192

The number of host bits is:

$$32 - 26 = 6$$

Step 2: Calculate the total number of IP addresses

$$2^6 = 64$$

Therefore, the subnet contains 64 total IP addresses.

Step 3: Determine the network address

The next available address after the first subnet is:

Network address: 192.168.100.128

The block size is 64. Hence, this subnet covers addresses from .128 through .191.

Step 4: Determine the broadcast address

The final address in the subnet is:

Broadcast address: 192.168.100.191

Step 5: Determine the valid host IP address range

The first usable host address is one greater than the network address. The last usable host address is one less than the broadcast address.

Therefore, the valid host range is:

192.168.100.129 – 192.168.100.190

Step 6: Calculate usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^6 - 2 \\ &= 64 - 2 \\ &= \boxed{62} \end{aligned}$$

Thus, the School of Electronics can connect up to 62 devices using this subnet.

School of Electronics – Subnet Details

Network address	192.168.100.128
Subnet mask	255.255.255.192
CIDR prefix	/26
Broadcast address	192.168.100.191
Valid host range	192.168.100.129 – 192.168.100.190
Total IP addresses	64
Usable host addresses	62

5. Subnet Allocation for the School of Mechanical Engineering (/27)

The School of Mechanical Engineering requires a subnet with the prefix length /27.

The first two departments use addresses up to 192.168.100.191. Therefore, the next available address is 192.168.100.192.

Step 1: Determine the subnet mask

The subnet mask for /27 is:

255.255.255.224

The number of host bits is:

$$32 - 27 = 5$$

Step 2: Calculate the total number of IP addresses

$$2^5 = 32$$

Therefore, this subnet contains 32 total IP addresses.

Step 3: Determine the network address

The next available address is:

Network address: 192.168.100.192

The block size is 32, so the subnet covers addresses from .192 through .223.

Step 4: Determine the broadcast address

The last address in the subnet is:

Broadcast address: 192.168.100.223

Step 5: Determine the valid host IP address range

The network address .192 and broadcast address .223 cannot be assigned to individual hosts.

Therefore, the valid host range is:

192.168.100.193 – 192.168.100.222

Step 6: Calculate usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^5 - 2 \\ &= 32 - 2 \\ &= \boxed{30} \end{aligned}$$

Thus, the School of Mechanical Engineering can connect up to 30 devices using this subnet.

School of Mechanical Engineering – Subnet Details

Network address	192.168.100.192
Subnet mask	255.255.255.224
CIDR prefix	/27
Broadcast address	192.168.100.223
Valid host range	192.168.100.193 – 192.168.100.222
Total IP addresses	32
Usable host addresses	30

6. Determine the Remaining Unused IP Address Range

The original network block is:

192.168.100.0/24

The complete address range is:

192.168.100.0 – 192.168.100.255

The three departments have been allocated the following address ranges:

- School of Computing: 192.168.100.0 – 192.168.100.127
- School of Electronics: 192.168.100.128 – 192.168.100.191
- School of Mechanical Engineering: 192.168.100.192 – 192.168.100.223

The last allocated address is 192.168.100.223.

The remaining addresses begin at 192.168.100.224 and continue through 192.168.100.255.

Therefore, the remaining unused IP address range is:

$$\boxed{192.168.100.224 - 192.168.100.255}$$

Calculation of remaining addresses

The original /24 block contains 256 addresses.

Total addresses allocated to the three departments:

$$128 + 64 + 32 = 224$$

Remaining addresses:

$$256 - 224 = \boxed{32}$$

Hence, 32 IP addresses remain unused.

These 32 addresses form a separate /27 subnet, which can be reserved for a future department, campus Wi-Fi network, laboratory, or other network expansion. If allocated as a conventional IPv4 subnet, it would provide 30 usable host addresses.

7. Complete Subnet Allocation Table

The following table summarizes the complete subnetting design.

Department	**Prefix**	**Subnet mask**	**Network address**	**Broadcast address**	**Valid host range**	**Usable hosts**
Computing	/25	255.255.255.128	192.168.100.0	192.168.100.127	.1 – .126	126
Electronics	/26	255.255.255.192	192.168.100.128	192.168.100.191	.129 – .190	62
Mechanical Engineering	/27	255.255.255.224	192.168.100.192	192.168.100.223	.193 – .222	30
Unused block	/27 equivalent	255.255.255.224	192.168.100.224	192.168.100.255	.225 – .254	30 if allocated

8. Benefits of This Subnetting Design

1. Efficient IP address utilization: Each department receives an appropriately sized subnet. Larger departments receive more addresses, while smaller departments use fewer addresses.

2. Improved network performance: Separating departments into subnets limits broadcast traffic within each subnet, reducing unnecessary traffic on other departmental networks.

3. Better network security: Access control lists and firewall rules can be applied between departmental subnets to restrict unauthorized communication and protect sensitive academic or administrative resources.

4. Simplified network administration: The administrator can identify devices based on their subnet, making troubleshooting, monitoring, and IP address management easier.

5. Future expansion: The remaining 32 addresses provide room for another subnet, allowing the Smart Campus network to expand without changing the existing departmental allocations.

Conclusion

Subnetting provides an organized and efficient method of dividing the Smart Campus network 192.168.100.0/24 into smaller departmental networks. The School of Computing receives 126 usable host addresses, the School of Electronics receives 62, and the School of Mechanical Engineering receives 30. After allocating the three subnets, the address range 192.168.100.224 – 192.168.100.255 remains unused, containing 32 IP addresses.

Thus, the subnet allocation satisfies all four requirements and provides a structured foundation for a secure, scalable, and manageable Smart Campus network.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Answer

Introduction

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. It is an important technique used in computer networking to improve IP address utilization, reduce broadcast traffic, simplify network administration, and enhance network security.

A multinational software company may have several departments, such as Software Development, Quality Assurance, Human Resources, and Executive Management. Each department requires a different number of IP addresses depending on the number of employees, computers, servers, printers, and other network devices.

By using subnetting, the network administrator can allocate different-sized subnets to each department according to its requirements.

In this problem, the company has been assigned the private IP address block 10.10.0.0/16. The following subnet masks have been allocated:

- Software Development – /24
- Quality Assurance – /25
- Human Resources – /26
- Executive Management – /27

The objective is to calculate the network address, broadcast address, valid host IP address range, number of usable hosts, and remaining unused IP address range for each subnet.

1. Understanding the Given IP Address Block

The given network address is:

10.10.0.0/16

The /16 prefix indicates that the first 16 bits represent the network portion, while the remaining 16 bits represent the host portion.

The subnet mask corresponding to /16 is:

255.255.0.0

The total number of IP addresses available in the block is calculated as follows:

$$\begin{aligned}\text{Total IP addresses} &= 2^{32-16} \\ &= 2^{16} \\ &= \boxed{65{,}536}\end{aligned}$$

Therefore, the complete address range is:

10.10.0.0 – 10.10.255.255

This block contains 65,536 IP addresses, which can be divided into smaller subnets for the company's departments.

2. Important Subnetting Formulas

The following formulas are used to calculate the subnet details.

1. Total number of IP addresses

$$\text{Total IPs} = 2^{32-\text{prefix length}}$$

2. Number of usable host addresses

$$\text{Usable hosts} = 2^{32-\text{prefix length}} - 2$$

Two addresses are normally reserved: one for the network address and one for the broadcast address.

3. Network address: The first address of a subnet, used to identify the network.

4. Broadcast address: The last address of a subnet, used to send packets to all hosts within that subnet.

5. Valid host range: The addresses between the network address and the broadcast address.

The subnets will be allocated consecutively, beginning with the lowest address in the given block. This ensures that the subnets do not overlap.

3. Subnet Allocation for Software Development (/24)

The Software Development department is assigned a /24 subnet. Since this department may contain many software developers, workstations, testing systems, and development servers, it receives the largest subnet among the four departments.

Step 1: Determine the subnet mask

The subnet mask for /24 is:

255.255.255.0

The number of host bits is:

$$32 - 24 = 8$$

Step 2: Calculate the total number of IP addresses

$$2^8 = 256$$

Therefore, the subnet contains 256 total IP addresses.

Step 3: Determine the network address

The first available network address is:

Network address: 10.10.0.0

This subnet covers addresses from 10.10.0.0 to 10.10.0.255.

Step 4: Determine the broadcast address

The last address in this subnet is the broadcast address.

Broadcast address: 10.10.0.255

Step 5: Determine the valid host IP address range

The network address and broadcast address cannot be assigned to regular hosts.

Therefore, the valid host range is:

10.10.0.1 – 10.10.0.254

Step 6: Calculate usable host addresses

$$\text{Usable hosts} = 2^8 - 2$$
$$= 256 - 2$$
$$= \boxed{254}$$

Thus, the Software Development department can accommodate up to 254 usable host addresses.

Software Development – Subnet Details

Network address	10.10.0.0
Subnet mask	255.255.255.0
Prefix length	/24
Broadcast address	10.10.0.255
Valid host range	10.10.0.1 – 10.10.0.254
Total IP addresses	256
Usable host addresses	254

4. Subnet Allocation for Quality Assurance (/25)

The Quality Assurance department is assigned a /25 subnet. This department uses computers and testing systems to check software functionality, identify bugs, and verify application performance.

Step 1: Determine the subnet mask

The subnet mask for /25 is:

255.255.255.128

The number of host bits is:

$$32 - 25 = 7$$

Step 2: Calculate the total number of IP addresses

$$2^7 = 128$$

Therefore, the subnet contains 128 total IP addresses.

Step 3: Determine the network address

The Software Development subnet occupies 10.10.0.0 – 10.10.0.255.

The next available address is 10.10.1.0.

Therefore:

Network address: 10.10.1.0

The /25 subnet covers addresses from 10.10.1.0 to 10.10.1.127.

Step 4: Determine the broadcast address

The last address in this subnet is:

Broadcast address: 10.10.1.127

Step 5: Determine the valid host IP address range

The first usable address is 10.10.1.1, and the last usable address is 10.10.1.126.

Therefore, the valid host range is:

10.10.1.1 – 10.10.1.126

Step 6: Calculate usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^7 - 2 \\ &= 128 - 2 \\ &= \boxed{126} \end{aligned}$$

Thus, the Quality Assurance department can accommodate 126 usable host addresses.

Quality Assurance – Subnet Details

Network address	10.10.1.0
Subnet mask	255.255.255.128
Prefix length	/25
Broadcast address	10.10.1.127
Valid host range	10.10.1.1 – 10.10.1.126
Total IP addresses	128
Usable host addresses	126

5. Subnet Allocation for Human Resources (/26)

The Human Resources department is assigned a /26 subnet. It generally requires IP addresses for employee computers, HR information systems, printers, and other administrative devices.

Step 1: Determine the subnet mask

The subnet mask for /26 is:

255.255.255.192

The number of host bits is:

$$32 - 26 = 6$$

Step 2: Calculate the total number of IP addresses

$$2^6 = 64$$

Therefore, the subnet contains 64 total IP addresses.

Step 3: Determine the network address

The Quality Assurance subnet occupies addresses up to 10.10.1.127.

The next available address is 10.10.1.128.

Therefore:

Network address: 10.10.1.128

The /26 subnet covers addresses from 10.10.1.128 to 10.10.1.191.

Step 4: Determine the broadcast address

The final address in this subnet is:

Broadcast address: 10.10.1.191

Step 5: Determine the valid host IP address range

The valid host range begins one address after the network address and ends one address before the broadcast address.

Therefore:

10.10.1.129 – 10.10.1.190

Step 6: Calculate usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^6 - 2 \\ &= 64 - 2 \\ &= \boxed{62} \end{aligned}$$

Thus, the Human Resources department can accommodate 62 usable host addresses.

Human Resources – Subnet Details

Network address	10.10.1.128
Subnet mask	255.255.255.192
Prefix length	/26
Broadcast address	10.10.1.191
Valid host range	10.10.1.129 – 10.10.1.190
Total IP addresses	64
Usable host addresses	62

6. Subnet Allocation for Executive Management (/27)

The Executive Management department is assigned a /27 subnet. Since this department generally has fewer network devices than the other departments, a smaller subnet is sufficient.

Step 1: Determine the subnet mask

The subnet mask for /27 is:

255.255.255.224

The number of host bits is:

$$32 - 27 = 5$$

Step 2: Calculate the total number of IP addresses

$$2^5 = 32$$

Therefore, the subnet contains 32 total IP addresses.

Step 3: Determine the network address

The Human Resources subnet occupies addresses up to 10.10.1.191.

The next available address is 10.10.1.192.

Therefore:

Network address: 10.10.1.192

The /27 subnet covers addresses from 10.10.1.192 to 10.10.1.223.

Step 4: Determine the broadcast address

The last address in this subnet is:

Broadcast address: 10.10.1.223

Step 5: Determine the valid host IP address range

The network address .192 and broadcast address .223 cannot be assigned to individual hosts.

Therefore, the valid host range is:

10.10.1.193 – 10.10.1.222

Step 6: Calculate usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^5 - 2 \\ &= 32 - 2 \\ &= \boxed{30} \end{aligned}$$

Thus, the Executive Management department can accommodate 30 usable host addresses.

Executive Management – Subnet Details

Network address	10.10.1.192
Subnet mask	255.255.255.224
Prefix length	/27
Broadcast address	10.10.1.223
Valid host range	10.10.1.193 – 10.10.1.222
Total IP addresses	32
Usable host addresses	30

7. Determine the Remaining Unused IP Address Range

The company has been assigned the network block:

10.10.0.0/16

The complete address range is:

10.10.0.0 – 10.10.255.255

This block contains 65,536 IP addresses.

The four departmental subnets are allocated consecutively as follows:

- Software Development: 10.10.0.0 – 10.10.0.255
- Quality Assurance: 10.10.1.0 – 10.10.1.127
- Human Resources: 10.10.1.128 – 10.10.1.191

- Executive Management: 10.10.1.192 – 10.10.1.223

The last allocated address is 10.10.1.223.

Therefore, the remaining unused address range starts at 10.10.1.224 and ends at 10.10.255.255.

Remaining unused IP address range:

$$\boxed{10.10.1.224 - 10.10.255.255}$$

Calculation of unused IP addresses

Total addresses allocated:

$$256 + 128 + 64 + 32 = 480$$

Total addresses in the original /16 block:

$$2^{32-16} = 65{,}536$$

Remaining unused addresses:

$$\text{Unused addresses} = 65{,}536 - 480$$
$$= \boxed{65{,}056}$$

Thus, 65,056 IP addresses remain unused after the four departmental subnet allocations.

This large remaining address range can be reserved for future departments, additional office floors, data centers, guest networks, or other network expansion. The unused range is not a single CIDR subnet because its starting point is 10.10.1.224 and its ending point is 10.10.255.255. It can be represented by multiple CIDR subnets if the administrator later decides to allocate it.

8. Complete Subnet Allocation Table

Department	Prefix	Subnet mask	Network address	Broadcast address	Valid host IP range	Usable hosts
Software Development	/24	255.255.255.0	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	/25	255.255.255.128	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	/26	255.255.255.192	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	/27	255.255.255.224	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

9. Benefits of the Subnet Allocation

1. Efficient IP address management: Different departments receive subnet sizes suited to their requirements, avoiding the need to assign every department an equally large network.

2. Improved network performance: Separating departmental networks helps limit broadcast traffic to the relevant subnet, improving overall network efficiency.

3. Enhanced security: Network administrators can apply access control lists and firewall policies between subnets. For example, access to sensitive HR and executive systems can be restricted to authorized users.

4. Easier troubleshooting: Since each department has its own subnet, administrators can identify the source of network issues more easily and manage devices systematically.

5. Scalability: The remaining 65,056 addresses provide substantial room for expansion. New subnets can be allocated to additional departments, branches, servers, and network services.

Conclusion

Subnetting allows the multinational software company to divide its private IP address block 10.10.0.0/16 into four departmental networks of different sizes. Software Development receives 254 usable host addresses, Quality Assurance receives 126, Human Resources receives 62, and Executive Management receives 30.

After allocating 480 total IP addresses to these four subnets, 65,056 IP addresses remain unused, from 10.10.1.224 to 10.10.255.255.

Therefore, the subnetting design meets all four requirements and provides an efficient, secure, and scalable network infrastructure for the company's present needs and future growth.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Answer

Introduction

Internet Protocol version 6 (IPv6) is the latest version of the Internet Protocol used to identify devices and enable communication across computer networks. It was developed to overcome the limitations of IPv4, particularly the shortage of available IP addresses.

A smart hospital uses thousands of connected devices, including patient monitoring systems, medical equipment, smart beds, IoT sensors, diagnostic machines, and hospital management systems. IPv6 is suitable for this environment because it provides an extremely large address space, supports efficient routing, and offers features that help with network configuration and communication.

In this scenario, the hospital has been assigned the following IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

The network engineer must convert the address into compressed notation, expand it into full hexadecimal representation, identify the network prefix, and calculate the number of available host addresses.

1. Convert the Given IPv6 Address into Compressed Notation

Definition of IPv6 address compression

An IPv6 address is a 128-bit address represented using eight groups of hexadecimal digits called hextets. Each hextet contains 16 bits and is represented by four hexadecimal digits separated by colons.

For example:

2001:0db8:abcd:0000:0000:0000:0000:0000

IPv6 compression reduces the length of an address by applying two rules:

Rule 1: Remove leading zeros.

Leading zeros in each hextet can be omitted without changing the value of the address.

Examples:

- 0db8 becomes db8.
- 0000 becomes 0.
- 0000 in any hextet can also be represented by an empty group when compressed using the second rule.

Rule 2: Replace consecutive zero hextets with double colons (::).

A continuous sequence of zero hextets can be replaced by ::. This notation can be used only once in an IPv6 address because using it more than once would make the number of omitted hextets ambiguous.

Step-by-step compression

The given address is:

2001:0db8:abcd:0000:0000:0000:0000:0000

First, remove the leading zeros from the hextets:

2001:db8:abcd:0:0:0:0:0

Next, identify the consecutive zero hextets at the end of the address. There are five consecutive zero hextets.

Replace those five zero hextets with ::.

Therefore, the compressed IPv6 address is:

`2001:db8:abcd::/64`

This is the shortest valid compressed notation for the given address while retaining the /64 network prefix.

2. Expand the Compressed Address into Full Hexadecimal Representation

Definition of IPv6 address expansion

IPv6 expansion is the process of converting a compressed IPv6 address into its complete representation. The expanded address must contain exactly eight hextets, with each hextet containing four hexadecimal digits.

The compressed address is:

2001:db8:abcd::/64

Step 1: Identify the existing hextets

The visible hextets are:

1. 2001
2. db8
3. abcd

There are three explicitly written hextets.

Step 2: Determine the number of omitted hextets

A complete IPv6 address contains eight hextets.

$$8 - 3 = 5$$

Therefore, the double colon (::) represents five zero hextets in this address.

Step 3: Restore the leading zeros

Expand db8 into four hexadecimal digits:

db8 becomes 0db8.

The zero hextets are written as 0000.

Step 4: Write the complete IPv6 address

The full hexadecimal representation is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

This expanded form contains eight hextets, each with four hexadecimal digits, making a total of 32 hexadecimal digits representing 128 bits.

Verification:

$$8 \times 16 = 128 \text{ bits}$$

Hence, the compressed and expanded forms represent the same IPv6 address.

3. Identify the Network Prefix Using the /64 Prefix Length

Definition of an IPv6 network prefix

The network prefix is the portion of an IPv6 address that identifies the network. The prefix length specifies the number of leading bits used for network identification.

In the given address, the prefix length is /64.

This means:

- The first 64 bits identify the network prefix.
- The remaining 64 bits are available for interface identification within the subnet.

An IPv6 address contains 128 bits in total.

$$\text{Interface identifier bits} = 128 - 64 = 64$$

Network prefix calculation

The given IPv6 address is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Since each hextet represents 16 bits, the first four hextets represent:

$$4 \times 16 = 64 \text{ bits}$$

Therefore, the network prefix consists of:

2001:0db8:abcd:0000

The network prefix in compressed notation is:

$$\boxed{2001{:}db8{:}abcd{:}0{::}/64}$$

The full address block can be represented as:

2001:db8:abcd:0:0000:0000:0000:0000/64

The prefix remains unchanged even when the host portion varies.

Importance of the /64 prefix

A /64 prefix is commonly used for IPv6 LANs because it provides a 64-bit interface identifier space and supports standard IPv6 Stateless Address Autoconfiguration (SLAAC) practices.

In the smart hospital, this prefix can be used to organize medical devices, monitoring equipment, and IoT sensors within a subnet. Additional networks can be created using different prefixes delegated from a larger IPv6 allocation.

4. Calculate the Total Number of Host Addresses Supported Within the Subnet

IPv6 uses 128-bit addresses. The number of addresses in a subnet depends on the number of bits remaining after the network prefix.

The given prefix is /64.

Step 1: Determine the number of interface identifier bits

$$128 - 64 = 64 \text{ bits}$$

Therefore, 64 bits are available for the interface identifier.

Step 2: Calculate the number of possible addresses

Each bit can have two values: 0 or 1.

The number of possible combinations of 64 bits is:

$$\boxed{2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616}$$

Therefore, the /64 subnet contains 18,446,744,073,709,551,616 possible IPv6 addresses.

Step 3: Understand the difference from IPv4

In conventional IPv4 subnet calculations, two addresses are generally excluded from host assignment: the network address and the broadcast address.

IPv6 does not use broadcast addresses. Instead, it uses multicast for communication with groups of interfaces.

Consequently, the IPv4 rule of subtracting two addresses should not be applied automatically to IPv6.

The /64 IPv6 subnet has 2^{64} possible interface identifier combinations, although practical address assignment depends on IPv6 configuration rules and interface requirements.

5. Determine the Total Number of Possible Host Addresses Available Within the Network

The fifth requirement asks for the total number of possible host addresses available in the assigned network. This is the same mathematical calculation as the fourth requirement, but it can be explained from the perspective of the entire subnet.

Given information

- IPv6 address: 2001:0db8:abcd:0000:0000:0000:0000:0000
- Prefix length: /64
- Total IPv6 address length: 128 bits
- Host/interface identifier portion: 64 bits

Calculation

The total number of address combinations is:

$$\begin{aligned}\text{Total addresses} &= 2^{128-64} \\ &= 2^{64} \\ &= \boxed{18{,}446{,}744{,}073{,}709{,}551{,}616}\end{aligned}$$

Thus, the network contains approximately 18.45 quintillion possible IPv6 addresses.

This enormous address capacity makes IPv6 highly suitable for smart hospitals, where a large number of connected devices may need unique addresses.

For example, the hospital could use addresses from the subnet for:

- Patient monitoring devices.
- Smart infusion pumps.
- Medical imaging equipment.
- IoT temperature and humidity sensors.
- Connected beds and wearable medical devices.
- Network printers, servers, and other infrastructure.

However, the number of usable interface addresses in practice depends on address assignment and device configuration. The mathematical capacity remains 2^{64}

6. Summary Table

Requirement	**Answer**
Given IPv6 address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Compressed notation	2001:db8:abcd::/64
Full hexadecimal representation	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Network prefix	2001:db8:abcd:0::/64
Prefix length	64 bits
Interface identifier length	64 bits
Total possible address combinations	2^{64}

Total numerical address capacity	18,446,744,073,709,551,616

7. Advantages of IPv6 in a Smart Hospital

1. Large address space: IPv6 provides an enormous number of addresses, enabling the hospital to connect thousands or millions of medical devices and IoT sensors.
2. Simplified network configuration: IPv6 supports Stateless Address Autoconfiguration (SLAAC), which allows compatible devices to configure their addresses automatically.
3. Improved routing efficiency: IPv6 uses a structured addressing hierarchy that can support scalable network design and route aggregation.
4. No broadcast traffic: IPv6 uses multicast and anycast mechanisms instead of broadcast, helping to reduce unnecessary network traffic.
5. Support for modern connected devices: IPv6 is well suited to environments with many connected sensors, medical instruments, and monitoring systems.
6. Security support: IPv6 supports IPsec as part of its protocol suite. However, secure deployment still requires appropriate authentication, firewall policies, access control, and network monitoring.

Conclusion

IPv6 is an effective addressing solution for the smart hospital because it provides a large address space and supports modern network configuration. The given address 2001:0db8:abcd:0000:0000:0000:0000:0000/64 can be compressed to 2001:db8:abcd::/64 and expanded back to its full hexadecimal representation.

The /64 prefix identifies the first 64 bits as the network prefix, leaving 64 bits for interface identification. Therefore, the subnet provides 2^{64}, or 18,446,744,073,709,551,616 possible IPv6 address combinations.

This capacity supports the hospital's growing requirements for medical equipment, IoT sensors, and connected healthcare systems, while providing a scalable foundation for future network expansion.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Answer

Introduction

IP routing is the process of selecting an appropriate path for forwarding data packets from a source network to a destination network. Routers operate at the Network Layer (Layer 3) of the OSI model and use routing tables to determine where packets should be forwarded.

An Internet Service Provider (ISP) connects multiple customer networks through core routers. These routers examine the destination IP address of each incoming packet and compare it with the entries in their routing tables. Based on the best matching entry, the router forwards the packet through the appropriate outgoing interface or next-hop router.

In this scenario, the ISP's core router maintains the following routing table entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

A packet arrives with the destination IP address 192.168.2.45.

The objective is to identify the destination subnet, matching routing entry, next-hop network/interface, and default route.

1. Determine the Destination Subnet to Which the Packet Belongs

Definition of a destination subnet

A subnet is a logical division of an IP network. Each subnet has a network address and a subnet mask that determine which IP addresses belong to that network.

The destination subnet is identified by comparing the destination IP address with the available network prefixes.

Given information

Destination IP address:

192.168.2.45

Available subnets:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Step 1: Understand the /24 prefix

A /24 prefix means that the first 24 bits identify the network and the remaining 8 bits identify the host.

The corresponding subnet mask is:

255.255.255.0

For a /24 network, the first three octets identify the network.

Step 2: Identify the destination network

The destination IP address is:

192.168.2.45

Its first three octets are:

192.168.2

Therefore, it belongs to the network:

192.168.2.0/24

Step 3: Verify the host range

The network 192.168.2.0/24 has the following address details:

- Network address: 192.168.2.0
- First usable host: 192.168.2.1
- Last usable host: 192.168.2.254
- Broadcast address: 192.168.2.255

The destination address 192.168.2.45 falls within the valid host range.

Therefore, the packet belongs to the 192.168.2.0/24 destination subnet.

2. Identify the Matching Routing Table Entry

Definition of a routing table

A routing table is a collection of network routes maintained by a router. Each entry identifies a destination network and provides forwarding information, such as an outgoing interface or next-hop router.

When a packet arrives, the router examines its destination IP address and searches for a matching routing entry.

Step 1: Compare the destination with each entry

The destination address is 192.168.2.45.

Routing table entry	**Does it match?**	**Explanation**
192.168.1.0/24	No	The third octet is 1, but the destination has 2.
192.168.2.0/24	Yes	The first three octets match the destination network.
192.168.3.0/24	No	The third octet is 3, but the destination has 2.

Step 2: Select the matching route

The routing table entry that matches the destination is:

`192.168.2.0/24`

Since the three entries have the same prefix length, there is no need to choose between them based on prefix specificity. Only the second entry matches the destination network.

Step 3: Apply the longest prefix match principle

Routers generally use the Longest Prefix Match (LPM) principle when multiple routing entries match a destination address.

This principle selects the matching route with the greatest number of prefix bits, because it represents the most specific route.

For example:

- 192.168.0.0/16 is a broader route.
- 192.168.2.0/24 is a more specific route.
- 192.168.2.0/25 is more specific still.

If several of these routes match the same destination, the router selects the matching route with the longest prefix.

In this scenario, only 192.168.2.0/24 among the supplied entries matches the destination, so it is selected.

3. Determine the Next-Hop Network or Interface Used to Forward the Packet

Definition of a next hop

A next hop is the next router or forwarding destination to which a packet is sent on its way to the final destination.

The outgoing interface is the router interface through which the packet leaves. Depending on the routing configuration, the route may specify a next-hop IP address, a directly connected interface, or both.

Step 1: Select the matching route

The selected routing entry is:

192.168.2.0/24

This identifies the destination network for forwarding.

Step 2: Determine the forwarding action

The router consults the forwarding information associated with this route.

There are two common possibilities.

Case 1: The destination network is directly connected.

If 192.168.2.0/24 is directly connected to the router, the router forwards the packet through the interface configured for that subnet.

For example, if the router has an interface named GigabitEthernet0/2 configured for 192.168.2.0/24, it can forward the packet through that interface. The actual interface name depends on the router configuration.

The router uses ARP to discover the destination device's MAC address when necessary on an Ethernet IPv4 network.

Case 2: The destination network is reachable through another router.

If the subnet is remote, the routing table must provide a next-hop router address or an equivalent forwarding instruction. The packet is forwarded to that next hop through the appropriate outgoing interface.

Step 3: State the result for this scenario

The destination network is:

192.168.2.0/24

The exact next-hop IP address or outgoing interface cannot be determined from the three network prefixes alone, because the question does not provide the associated next-hop addresses or interface information.

Therefore, the correct routing decision is to use the forwarding information associated with the 192.168.2.0/24 route.

4. Determine the Default Route When No Entry Matches

Definition of a default route

A default route is a route used when the destination IP address does not match any more specific route in the routing table.

It acts as a fallback path and is particularly useful when a router forwards unknown destinations toward an upstream router or an ISP gateway.

The IPv4 default route is represented as:

0.0.0.0/0

The /0 prefix means that no bits are required to match a destination. Consequently, the default route can match any IPv4 destination, provided no more specific applicable route takes precedence.

Step 1: Understand the default route

The default route is:

$$\boxed{0.0.0.0/0}$$

It is commonly configured with a next-hop IP address belonging to an upstream router.

For example, the following is an illustrative route configuration:

0.0.0.0/0 via 192.168.1.1

Here, 192.168.1.1 is only an example of a possible next hop; it is not specified in the question.

Step 2: Explain how the router uses it

Suppose a packet arrives with a destination address of 172.16.5.10.

This destination does not match any of the three supplied /24 routes:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

If the router has a configured default route, it forwards the packet using that route's next-hop or outgoing interface.

If no default route exists and no other route matches, the router generally drops the packet and may send an ICMP Destination Unreachable message, depending on its configuration.

Step 3: State the default route

The default route to use when no more specific route matches is:

0.0.0.0/0

The actual next-hop IP address or outgoing interface must be obtained from the router's configuration.

5. Complete Routing Decision Table

Requirement	Answer
Destination IP address	192.168.2.45
Destination subnet	192.168.2.0/24
Matching routing table entry	192.168.2.0/24
Subnet mask	255.255.255.0
Network address	192.168.2.0
Broadcast address	192.168.2.255
Valid host range	192.168.2.1 – 192.168.2.254
Next-hop network	192.168.2.0/24
Exact next-hop/interface	Depends on the route's forwarding information
Default route	0.0.0.0/0

6. Importance of Routing in an ISP Network

1. Efficient packet forwarding: Routing tables allow routers to select appropriate paths for packets traveling between customer networks.

2. Network scalability: ISPs can connect many customer networks while maintaining structured routing information.

3. Reduced unnecessary traffic: Correct route selection helps deliver packets to their intended destinations rather than forwarding them along unrelated paths.

4. Reliability: Alternative routes and suitable routing protocols can help maintain connectivity when a link or router becomes unavailable.

5. Flexible network management: Administrators can configure static routes, dynamic routing protocols, and default routes according to the requirements of the network.

Conclusion

IP routing enables an ISP's core router to forward packets toward the correct destination network by examining destination IP addresses and consulting its routing table. For the destination IP address 192.168.2.45, the matching subnet is

192.168.2.0/24, and the corresponding routing table entry is selected for forwarding.

The exact next-hop IP address or outgoing interface cannot be identified without additional routing information. If no specific route matches another packet's destination, the router can use the default route 0.0.0.0/0, provided that route is configured.

Thus, routing tables, longest prefix matching, and default routes are essential mechanisms for efficient and reliable packet forwarding in ISP networks.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Answer

Introduction

A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area, such as an office, building, or campus. In an organization, different departments may require separate LANs to improve network performance, security, and administration.

A Layer-3 router connects different IP networks and enables communication between them. It operates at the Network Layer of the OSI model and forwards packets based on destination IP addresses and its routing table.

In this scenario, a company has two departments, Administration and Finance, connected through a Layer-3 router. Each department has been assigned a separate subnet using the IPv4 private address range 192.168.10.0/24.

The subnet allocations and router gateway addresses are:

- Administration LAN: 192.168.10.0/26
- Finance LAN: 192.168.10.64/26
- Administration Gateway: 192.168.10.1
- Finance Gateway: 192.168.10.65

The objective is to determine the network and broadcast addresses, identify the valid host ranges, suggest IP addresses for three computers in each LAN, and identify the default gateway used by hosts.

1. Understanding the Given Subnet Information

Before calculating the network details, it is important to understand the /26 prefix.

The subnet mask corresponding to /26 is:

255.255.255.192

An IPv4 address contains 32 bits. The /26 prefix indicates that 26 bits are used for the network portion and the remaining 6 bits are used for the host portion.

The number of host bits is:

$$32 - 26 = 6$$

The total number of IP addresses in each subnet is:

$$2^6 = 64$$

The number of usable host addresses is:

$$64 - 2 = 62$$

Two addresses are normally reserved: the network address and the broadcast address.

Therefore, each LAN supports 64 total IP addresses and 62 usable host addresses.

The block size can also be calculated using the last octet of the subnet mask:

$$256 - 192 = 64$$

Hence, the subnets increase in blocks of 64 addresses in the last octet.

The first subnet starts at 192.168.10.0, and the second starts at 192.168.10.64.

2. Determine the Network Address and Broadcast Address for Each LAN

A. Administration LAN – 192.168.10.0/26

Step 1: Identify the network address

The Administration LAN is assigned the subnet:

192.168.10.0/26

The network address is the first address in the subnet.

Therefore:

Network address: 192.168.10.0

Step 2: Calculate the broadcast address

The subnet contains 64 addresses, beginning at .0.

The final address is:

$$0 + 64 - 1 = 63$$

Therefore:

Broadcast address: 192.168.10.63

The broadcast address is used to send packets to all devices within the subnet.

Step 3: Determine the valid host range

The first usable host address is one greater than the network address:

192.168.10.1

The last usable host address is one less than the broadcast address:

192.168.10.62

Therefore, the valid host range is:

192.168.10.1 – 192.168.10.62

Step 4: Calculate the usable host addresses

$$\begin{aligned} \text{Usable hosts} &= 2^{32-26} - 2 \\ &= 64 - 2 \\ &= \boxed{62} \end{aligned}$$

Thus, the Administration LAN can accommodate up to 62 hosts.

Administration LAN – Address Details

Network address	192.168.10.0
Subnet mask	255.255.255.192
Prefix length	/26
Broadcast address	192.168.10.63
Valid host range	192.168.10.1 – 192.168.10.62
Total addresses	64
Usable hosts	62

B. Finance LAN – 192.168.10.64/26

Step 1: Identify the network address

The Finance LAN is assigned:

192.168.10.64/26

The second subnet begins at .64, which is the next valid network boundary for a /26 subnet.

Therefore:

Network address: 192.168.10.64

Step 2: Calculate the broadcast address

The subnet contains 64 addresses, beginning at .64.

The final address is:

$$64 + 64 - 1 = 127$$

Therefore:

Broadcast address: 192.168.10.127

Step 3: Determine the valid host range

The first usable host address is:

192.168.10.65

The last usable host address is:

192.168.10.126

Therefore, the valid host range is:

192.168.10.65 – 192.168.10.126

Step 4: Calculate the usable host addresses

$$\begin{aligned}\text{Usable hosts} &= 2^6 - 2 \\ &= 64 - 2 \\ &= \boxed{62}\end{aligned}$$

Thus, the Finance LAN can accommodate up to 62 hosts.

Finance LAN – Address Details

Network address	192.168.10.64
Subnet mask	255.255.255.192
Prefix length	/26
Broadcast address	192.168.10.127
Valid host range	192.168.10.65 – 192.168.10.126
Total addresses	64
Usable hosts	62

3. Suggest Suitable IP Addresses for Three Computers in Each LAN

Each computer must be assigned a unique IP address within its department's valid host range. The network address and broadcast address cannot be assigned to regular computers. The gateway address must also be reserved for the router interface to avoid IP address conflicts.

A. IP address allocation for Administration LAN

The Administration LAN has the valid host range:

192.168.10.1 – 192.168.10.62

The router gateway is 192.168.10.1. Therefore, this address is reserved for the router and should not be assigned to a computer.

Three suitable computer addresses are:

Device	**Assigned IP address**	**Subnet mask**	**Default gateway**
Administration PC 1	192.168.10.2	255.255.255.192	192.168.10.1
Administration PC 2	192.168.10.3	255.255.255.192	192.168.10.1

Administration PC 3	192.168.10.4	255.255.255.192	192.168.10.1

These addresses belong to the Administration subnet and do not conflict with the router gateway.

B. IP address allocation for Finance LAN

The Finance LAN has the valid host range:

192.168.10.65 – 192.168.10.126

The router gateway is 192.168.10.65. Therefore, this address is reserved for the router interface.

Three suitable computer addresses are:

Device	Assigned IP address	Subnet mask	Default gateway
Finance PC 1	192.168.10.66	255.255.255.192	192.168.10.65
Finance PC 2	192.168.10.67	255.255.255.192	192.168.10.65
Finance PC 3	192.168.10.68	255.255.255.192	192.168.10.65

These addresses are within the Finance subnet and are different from the gateway address.

Importance of proper IP address allocation

1. Prevents address conflicts: Each computer receives a unique IP address.
2. Ensures correct subnet membership: Computers are assigned addresses belonging to their respective LANs.
3. Avoids gateway conflicts: The router's interface addresses are reserved and are not assigned to computers.
4. Simplifies administration: Sequential addresses make device identification and troubleshooting easier.
5. Supports communication: Correct subnet masks and gateway settings allow computers to communicate with local and remote networks.

4. Identify the Default Gateway for Hosts in Each LAN

Definition of a default gateway

A default gateway is the router interface that a computer uses to send packets to destinations outside its own local subnet.

When a computer communicates with another device in the same subnet, it can send packets directly to that device through the LAN. However, when the destination belongs to a different network, the computer forwards the packet to its configured default gateway.

The Layer-3 router then examines the destination IP address and determines the appropriate outgoing interface.

A. Default gateway for Administration LAN

The given Administration gateway is:

192.168.10.1

All computers in the Administration LAN should use 192.168.10.1 as their default gateway.

For example, if Administration PC 1 (192.168.10.2) needs to communicate with Finance PC 1 (192.168.10.66), the destination belongs to a different subnet.

Therefore, Administration PC 1 sends the packet to 192.168.10.1. The router then forwards it through its Finance LAN interface toward 192.168.10.66.

B. Default gateway for Finance LAN

The given Finance gateway is:

192.168.10.65

All computers in the Finance LAN should use 192.168.10.65 as their default gateway.

For example, if Finance PC 1 (192.168.10.66) needs to communicate with Administration PC 1 (192.168.10.2), the destination is outside the Finance subnet. Therefore, Finance PC 1 sends the packet to 192.168.10.65. The router then forwards it through its Administration LAN interface toward 192.168.10.2.

Summary of default gateways

Department	Default gateway	Purpose
Administration	192.168.10.1	Forwards packets from Administration to other networks
Finance	192.168.10.65	Forwards packets from Finance to other networks

The default gateway must belong to the same subnet as the host using it. Both configured gateways satisfy this requirement.

5. How the Layer-3 Router Enables Communication Between the LANs

A Layer-3 router connects the Administration and Finance subnets and enables inter-subnet communication.

A simplified network representation is:

Administration LAN

PCs: 192.168.10.2, .3, .4

Gateway: 192.168.10.1/26

Layer-3 Router

Routes packets between both subnets

Finance LAN

PCs: 192.168.10.66, .67, .68

Gateway: 192.168.10.65/26

Steps involved in inter-subnet communication

Step 1: Destination identification

When a computer sends data, it compares the destination IP address with its own IP address and subnet mask to determine whether the destination is local or remote.

Step 2: Forwarding to the gateway

If the destination belongs to another subnet, the computer sends the packet to its configured default gateway.

Step 3: Routing table lookup

The router checks its routing table and identifies the outgoing interface corresponding to the destination network.

Step 4: Packet forwarding

The router forwards the packet through the appropriate interface. For example, a packet from Administration to Finance is forwarded through the Finance interface.

Step 5: Delivery to the destination

The packet reaches the destination LAN, where it is delivered to the intended computer.

This mechanism enables the two departments to communicate while maintaining separate IP subnets.

6. Complete Subnet Calculation Summary

Parameter	**Administration LAN**	**Finance LAN**
Network address	192.168.10.0	192.168.10.64
CIDR prefix	/26	/26
Subnet mask	255.255.255.192	255.255.255.192
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Total IP addresses	64	64
Usable host addresses	62	62
Router gateway	192.168.10.1	192.168.10.65
Three example PCs	.2, .3, .4	.66, .67, .68

7. Advantages of This Network Design

1. Improved security: Separate subnets make it easier to implement firewall rules and access control policies between Administration and Finance.

2. Efficient network management: Each department has a distinct network address range, allowing administrators to identify and manage devices more easily.
3. Reduced broadcast traffic: Broadcast traffic is generally confined to its own subnet, reducing unnecessary traffic across the other LAN.
4. Centralized inter-network routing: The Layer-3 router provides a central point for routing traffic between the two departments.
5. Scalability: Each subnet supports up to 62 usable hosts, allowing additional computers and devices to be added within the available address range.

Conclusion

Subnetting and Layer-3 routing allow a company to connect multiple departments efficiently while keeping their networks logically separate. In this scenario, the Administration LAN uses 192.168.10.0/26, with broadcast address 192.168.10.63 and 62 usable host addresses. The Finance LAN uses 192.168.10.64/26, with broadcast address 192.168.10.127 and 62 usable host addresses.

The Administration computers use 192.168.10.1 as their default gateway, while the Finance computers use 192.168.10.65. By assigning unique IP addresses to each computer and configuring the correct subnet mask and gateway, the organization can enable reliable communication between departments.

Thus, the proposed network configuration meets all four requirements and provides an organized, manageable, and scalable network infrastructure for the company.